Study Questions: Systems

1. When you hold your crochet work-in-progress, identify the **internal states**, **external states**, and **blanket states**. What counts as "inside" the system and what counts as "outside"?

2. The live loop on your hook is described as the core of the Markov blanket in crochet. Why is this loop so critical — what happens to the system if it slips off the hook?

3. Compare a crochet project that is actively being worked to one that is set down and stored in a bag. In what sense does the stored project still have a Markov blanket? In what sense has the boundary changed?

4. How does the tension on your working yarn serve as a **sensory state** — a channel through which external information reaches the internal system?

5. When you crochet into the top loops of the previous row, you are interacting with internal states through the boundary. Explain how this relates to the concept of conditional independence in a Markov blanket.

6. A crochet gauge swatch is made before the main project begins. From a systems perspective, what role does the swatch play in establishing the relationship between internal and external states?

7. Describe what happens to the system boundary during **frogging** (ripping back stitches). Which states change their classification from internal to external?

8. Your yarn supply gradually diminishes as you crochet. How does this change in external states affect the system as a whole? At what point might it threaten the system's integrity?

9. If you switch yarn colors mid-row, you are introducing a new external state into the system. How does this transition happen at the Markov blanket? What does the crocheter need to manage at the boundary?

10. Consider two crocheters sitting side by side, each working on their own project. Are they one system or two? Where would you draw the Markov blanket(s)?

11. A crochet pattern written on paper is an external state. How does the information in the pattern cross the Markov blanket to influence the internal states of the fabric?

12. In Active Inference, the Markov blanket is defined by statistical conditional independence. Explain this concept using the example of a completed row of single crochet stitches and the ball of yarn in the skein.

13. How does the size of your crochet hook relate to the properties of the Markov blanket? What changes about the boundary when you switch from a 4mm to a 6mm hook?

14. When crocheting in the round (such as making a hat), the working edge forms a continuous loop rather than a row. How does this change the topology of the Markov blanket compared to flat, row-by-row crochet?

15. A stitch marker placed at the beginning of a round serves as a reference point. Is the stitch marker part of the internal states, external states, or blanket states of the system? Justify your answer.

16. Consider the concept of **yarn weight** (lace, sport, worsted, bulky). How does yarn weight function as an external state constraint on the system? How does it shape what internal states are possible?

17. When you join a crochet circle and share a pattern with others, the external states of your system expand. How does the social environment of the circle interact with your individual crochet system?

18. In Active Inference, systems tend to resist dissolution — they maintain their Markov blanket over time. How does a crocheter actively maintain the integrity of their project's boundary? What habits or practices serve this function?

19. If you pick up someone else's unfinished project and continue it, are you working within the same system or creating a new one? How do you account for the change in agent while the internal states (existing fabric) remain?

20. Reflect on the entire lifecycle of a crochet project — from winding the yarn to weaving in the final ends. At what point does the system "come into existence," and at what point does it cease to be an active system? What remains when the project is complete?